

TARAPRASAD SHADANGI

Senior Embedded Software Engineer | Firmware Development

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SUMMARY

Embedded Software Engineer professional with over 3 years of experience in embedded software development and automotive project management, expert in DFMEA and firmware development. My key achievements include the successful management of complex automotive projects adhering to ISO and IEC standards and the consistent delivery of firmware meeting TUV SUD certification requirements. Seeking an Embedded Developer position, where I bring my embedded software development and automotive project management skills to support your mission of innovation in technology solutions.

EXPERIENCE

TRAINEE ENGINEER

elnfochips

01/2022 - 08/2022 Pune, Maharashtra

elnfochips is a technology company specializing in embedded solutions

- Innovated solutions for complex engineering challenges, resulting in 30% reduction in project turnaround time
- Participated in 20 failure analyses, improving system reliability through thorough report documentation
- Implemented procedures ensuring 100% compliance with engineering standards, enhancing product quality
- Improved manufacturing efficiency by analyzing workflows, leading to 10% increase in production speed

EMBEDDED ENGINEER

elnfochips

08/2022 - 04/2025 Pune, Maharashtra

elnfochips is a technology company specializing in embedded solutions

- Contributed to testing of embedded software for 7 microcontrollers, ensuring robust product performance
- Implemented advanced technologies in 2 major projects by continuously updating technical knowledge through workshops
- Performed diagnostics on 5 systems, ensuring 98% operational reliability across all platforms
- Developed complete embedded solutions with 25 engineers, fostering innovative project collaboration
- Reduced project delivery time by 15% by automating repetitive development tasks
- Worked collaboratively on a team of 10 engineers, covering projects with a combined total of around 100,000 lines of code

SENIOR EMBEDDED SOFTWARE ENGINEER

elnfochips

04/2025 - Present Pune, Maharashtra

elnfochips is a technology company specializing in embedded solutions

- Participated in Design Failure Mode and Effect Analysis (DFMEA) for automotive projects, ensuring compliance with ISO and relevant IEC 61851 standards
- Delivered firmware for 2 products, meeting 100% requirements for TUV SUD certification processes
- Successfully managed and executed board bring-up activities of 3 products for hardware validation
- Mentored 5 junior engineers on best practices in embedded systems development, fostering culture of continuous learning and improvement
- Reduced firmware update time by 47%, resulting in increased production efficiency by optimizing embedded code pathways

KEY ACHIEVEMENTS



Automotive Project Management and Mentorship

Successful management of complex automotive projects that adhere to stringent ISO and IEC standards, as well as my ongoing commitment to mentoring junior engineers in embedded system development



TUV SUD Certification

Consistent delivery of firmware that meets the requirements for TUV SUD certification

SKILLS

Data Structures	DFMEA	Firmware
FreeRTOS	I2C	UART
SPI	STM32	
TIMSPM0	NXPS32K344	C Language
RISC V Architecture	Automotive Standards	
Scripting		

TOOLS

Logic Analyzer	STM Cube IDE
Code Composer Studio	NXP Design Studio
Oscilloscope	Lin Analyzer
Can Analyzer	
JTAG Debugger	Static Code Analyzer

PROJECTS

EV CHARGER(MODE 2)

A project centered on developing a Single Phase AC electric vehicle (EV) charger tailored for the North American market

- Developed a Single Phase AC electric vehicle (EV) charger ensuring compliance with the SAE J1772 standard
- Implemented ADC, GPIO, Free RTOS, UART, and SPI in the development
- Utilized STM32G473VETx microcontroller

EDUCATION

B.TECH IN ELECTRONICS AND COMMUNICATION

MIT ADT - School of Engineering

08/2018 - 05/2022 Pune

PUBLICATIONS

Drone (Unmanned Aerial Vehicle) for Surveillance

IJISRT

05/2020 <https://www.ijisrt.com/drone-unmanned-aerial-vehicle-for-surveillance>

Autonomous Vehicle System

IJRASET

06/2022 <https://doi.org/10.22214/ijraset.2022.44322>

AWARDS



Academic excellence award



Power Trainee



Core Value Award

PROJECTS

EV CHARGER(MODE 3)

This project is focused on developing a Mode 3 AC electric vehicle (EV) charging station for the European market

- Engineered a Mode 3 AC electric vehicle (EV) charging station for the European market
- Supported Single Phase and Three Phase power inputs and compliance with the IEC 61851 standard
- Executed ADC, GPIO, FreeRTOS, UART, SPI, Watchdog, firmware updates, and SHA-256 functionalities
- Utilized STM32G484QETx microcontroller

SOFTWARE DEFINED VEHICLE

A proof-of-concept project demonstrating Software-Defined Vehicle principles

- Demonstrated Software-Defined Vehicle (SDV) principles using an ESP32 microcontroller
- Managed connected hardware features based on subscription status via vehicle networks
- Employed GPIO, UART, and CAN technologies

FIRMWARE MIGRATION

A firmware migration project focused on adapting existing applications to new microcontroller platforms

- Migrated a sensing application from an STM8 microcontroller to MSPM0L1306
- Adapted firmware for 20 distinct hardware architectures, ensuring functional equivalence across all systems
- Worked on board bring-up and sensor data transmission via LIN